Rethinking the policy optimization with Gradient Estimators

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Abstract

1 Introduction

Reinforcement learning has become central to the post-training of large language models (LLMs) [1, 15, 18]. In scalable training systems, policy optimization is often effectively off-policy: a rollout policy generates responses, while a trainer reuses the collected data for one or more gradient updates. Consequently, the current policy need not coincide with the policy that generated the data [11]. This mismatch makes it important to control how far policy optimization moves from the rollout policy.

Controlling such policy deviation has long been a central principle in policy optimization. Trust Region Policy Optimization (TRPO) formalizes this principle through population-level policy improvement guarantees, while Proximal Policy Optimization (PPO) provides a simple and scalable approximation through ratio-based clipping [13, 14]. PPO-style objectives, including GRPO, inherit this intuition by clipping or masking updates when sampled likelihood ratios move too far from one [15]. The resulting theoretical picture is clear under population view: if the population gradient estimators available, keeping the updated policy sufficiently close to the rollout policy controls the error of the surrogate objective and thereby protects policy improvement.

LLM training, however, presents a striking tension with this population-level view. Empirically, permissive estimators-including truncated importance sampling (TIS) with large caps or even unclipped updates-can remain stable and learn considerably faster. Yet this success need not persist. Figure 1 illustrates this phenomenon: TIS with caps 3 and 5 improves steadily for many updates before abruptly collapsing. The same update rule can therefore be highly effective for a long period and suddenly become unreliable. This raises the central question of this paper: *when does a permissive policy-gradient estimator cease to provide a reliable update?*

Classical trust-region theory explains why moving too far from the rollout policy can be unsafe [11, 13, 14]. It does not, however, explain the delayed transition in Figure 1, where failure follows many apparently successful updates rather than a single conspicuous policy shift. Repeated updates can progressively concentrate complete-trajectory importance weights, reducing the effective support provided by a finite rollout even when the empirical objective is constructed from factorwise coefficients. A population-level bound controls policy deviation, but it does not characterize whether the available rollout still contains enough effective information to estimate the next update reliably. The relevant question is therefore not only *how far the policy has moved*, but also *whether the finite rollout still supports a reliable gradient estimate.*

In this paper, we resolve this tension by shifting the analysis from population-level policy deviation to finite-sample update reliability. We show that the reliability of the policy-gradient

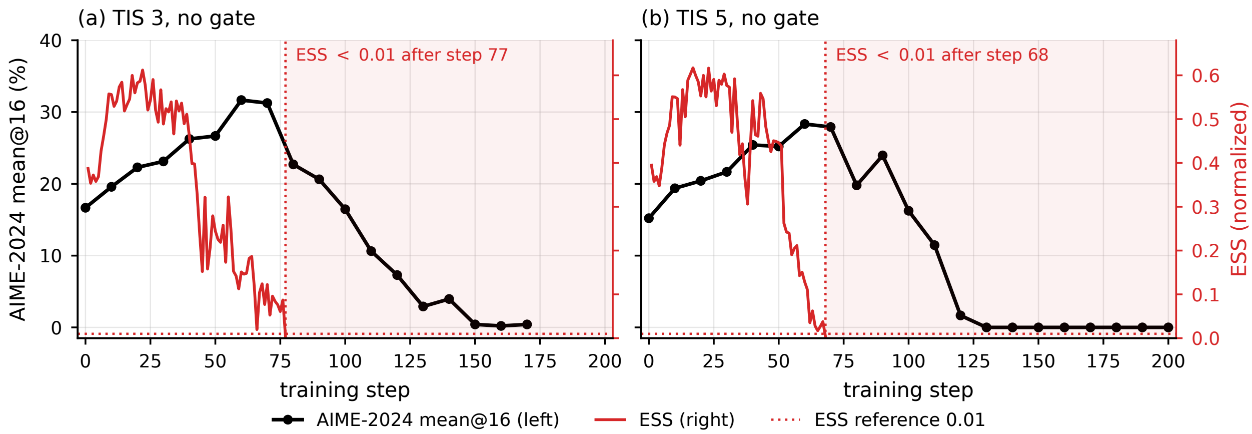

Figure 1: Delayed failure under permissive TIS updates on Qwen3-30B-A3B. TIS with caps 3 and 5 improves AIME-2024 mean@16 for many updates before effective sample size (ESS) deteriorates and performance subsequently collapses. The delayed transition motivates our central question: when does a finite rollout cease to support a reliable policy-gradient update?

estimator plays a central role in determining whether an empirical update translates into policy improvement. This perspective recasts clipping as a form of estimator control: permissive estimators can preserve more of the learning signal when the rollout provides sufficient effective support, whereas controlling the estimator becomes increasingly valuable as that support deteriorates. We make this connection explicit through effective sample size (ESS), which quantifies the concentration of the importance weights underlying the finite-sample estimator. We then validate this prediction empirically and use it to derive a simple optimization rule that preserves permissive updates while they remain reliable and intervenes when the rollout ceases to support them.

In this work, we develop a finite-sample perspective on policy optimization that explains when permissive updates remain reliable and when stronger control becomes necessary. Our contributions are threefold. **Finite-Sample Policy Improvement:** We derive a policy-improvement analysis that explicitly accounts for the reliability of the empirical gradient estimator, complementing the population-level view of policy deviation with the statistical adequacy of the available rollout. **Estimator Reliability and ESS:** We interpret clipping and truncation as forms of estimator control and establish a direct connection between gradient-estimator reliability and effective sample size (ESS), explaining why permissive updates can learn rapidly when effective support is sufficient but become unstable as importance weights concentrate. **Reliability-Aware Optimization:** Guided by this analysis, we propose a simple optimization rule that preserves permissive updates while the rollout remains reliable and increases control only when reliability deteriorates. Controlled simulations and large-scale LLM reasoning experiments validate these predictions and demonstrate an improved stability-learning tradeoff over fixed clipping strategies.

2 Related work

Policy-deviation guarantees. Trust-region and proximal methods control policy movement to obtain stable improvement guarantees or practical update rules [11, 13, 14]. These analyses provide the standard explanation for why large policy changes can be unsafe. Our framework is complementary: it studies when a fixed finite rollout ceases to provide a reliable estimate of the current policy gradient. The distinction is between a population-level approximation guarantee and a finite-sample reliability question.

Importance sampling and effective support. Importance-sampling policy optimization uses likelihood-ratio moments and Rényi divergence to quantify the reliability of off-policy estimators [5, 6]. In these results, effective rather than nominal sample size controls estimation error. We apply the same principle directly to the vector-valued policy-gradient estimator and use normalized ESS as the regime coordinate.

Reliable policy-gradient updates. Smoothness-based analyses translate the error of a finite gradient estimate into a lower bound on policy improvement and use this relation to choose step sizes or sample sizes [8, 10]. We combine this optimization argument with an ESS-dependent off-policy reliability bound.

Stability in RLVR. Recent methods alter ratio boundaries, sequence weights, divergence controls, or update rules to improve the stability of large-model reinforcement learning [2, 3, 7, 9, 16, 18, 19]. Rather than proposing another static boundary, we study the statistical transition that makes a permissive update unreliable and then ask when an intervention can recover reliability.

3 Preliminaries

This section develops the estimator formulation used throughout our analysis. For a fixed rollout policy, change of measure expresses the off-policy policy gradient as an importance-weighted population mean. This formulation places importance sampling (IS) and truncated importance sampling (TIS) within the class of importance-weighted estimators, while distinguishing PPO/GRPO clipping, which masks selected gradient contributions. We then introduce effective sample size (ESS) as the second-moment quantity used to characterize the finite-sample reliability of importance weighting.

3.1 Policy gradients as importance-weighted estimators

Let $X \sim \nu$ denote a context, $Q(\cdot \mid X)$ the rollout policy, and $P_\theta(\cdot \mid X)$ the current policy. A complete sample is $Z = (X, Y)$, where $Y = (Y_1, \dots, Y_T)$ and $R(Z)$ is its reward. For each factor t, define the likelihood ratio and score

$$r_{\theta,t}(Z) := \frac{P_\theta(Y_t \mid X, Y_{<t})}{Q(Y_t \mid X, Y_{<t})}, \qquad \ell_{\theta,t}(Z) := \nabla_\theta \log P_\theta(Y_t \mid X, Y_{<t}). \tag{1}$$

Here t indexes a policy factor, such as a token or decision step. Since both policies factorize along the trajectory, the complete-sample likelihood ratio is

$$W_\theta(Z) := \frac{P_\theta(Y \mid X)}{Q(Y \mid X)} = \prod_{t=1}^{T} r_{\theta,t}(Z). \tag{2}$$

The learning objective is to maximize the expected reward $J(\theta) := \mathbb{E}_{P_\theta}[R]$. Let $A(Z) := R(Z) - b(X)$ denote a baseline-centered reward, where $b(X)$ does not depend on Y. Under $P_\theta(\cdot \mid X) \ll Q(\cdot \mid X)$, change of measure expresses the population gradient under the rollout distribution:

$$g(\theta) := \nabla J(\theta) = \mathbb{E}_Q[W_\theta f_\theta], \qquad f_\theta(Z) := A(Z) \sum_{t=1}^{T} \ell_{\theta,t}(Z) = A(Z) \nabla_\theta \log P_\theta(Y \mid X). \tag{3}$$

Equation (3) identifies off-policy policy-gradient estimation as an importance-weighted mean-estimation problem. The population quantity $g(\theta)$ is not observed directly. Given i.i.d. rollout samples $Z_i \sim \nu(dx)Q(dy \mid x)$, $i = 1, \dots, N$, its full-trajectory IS estimator is

$$\widehat{g}_N(\theta) := \frac{1}{N} \sum_{i=1}^{N} W_{\theta,i} f_{\theta,i}. \tag{4}$$

The policy update therefore depends on how accurately this finite importance-weighted mean estimates its population target.

Practical policy-gradient methods differ in how they modify the coefficients entering this empirical mean. To represent these choices in a common form, write $\mathbf{r}_\theta = (r_{\theta,1}, \dots, r_{\theta,T})$ and let $u = (u_1, \dots, u_T)$ be a collection of coefficient rules, with $u_t : \mathbb{R}_+^T \times \mathbb{R} \to \mathbb{R}$. Define

$$\psi_\theta(u; Z) := A(Z) \sum_{t=1}^{T} u_t(\mathbf{r}_\theta, A(Z)) \ell_{\theta,t}(Z), \qquad \widehat{g}_N(u; \theta) := \frac{1}{N} \sum_{i=1}^{N} \psi_\theta(u; Z_i). \tag{5}$$

Full-trajectory IS uses

$$u_{\mathrm{IS},t} = W_\theta \qquad \text{for every } t. \tag{6}$$

A factorwise TIS rule instead caps the importance coefficient,

$$u_{\mathrm{T},t}(\mathbf{r}, A) := \min\{r_t, c\}, \tag{7}$$

while retaining the corresponding score contribution. PPO/GRPO ratio clipping operates differently: the clipped surrogate induces the advantage-dependent gradient-masking rule [14, 15]

$$u_{\mathrm{P},t}(\mathbf{r}, A) := r_t \left[\mathbf{1}\{A \geq 0,\ r_t \leq 1 + \epsilon\} + \mathbf{1}\{A < 0,\ r_t \geq 1 - \epsilon\}\right], \qquad 0 < \epsilon < 1. \tag{8}$$

IS and TIS are both importance-weighted estimators: truncation changes the magnitude of selected importance coefficients while retaining their gradient contributions. PPO/GRPO clipping instead removes selected contributions once the clipping condition is active. This distinction leads to two questions that motivate the theoretical analysis. First, when is an importance-weighted policy-gradient estimator reliable at finite sample size? Second, when does the reduction in sampling variation from clipping compensate for the gradient signal removed by masking?

3.2 Importance-weight concentration and effective sample size

The finite-sample behavior of an importance-weighted mean depends on the dispersion of its likelihood weights [5, 6]. Rényi divergence provides a second-moment characterization of this dispersion. For $\alpha > 1$, define

$$D_\alpha(P_\theta \| Q) := \frac{1}{\alpha - 1} \log \mathbb{E}_Q[W_\theta^\alpha], \qquad d_\alpha(P_\theta \| Q) := \exp\{D_\alpha(P_\theta \| Q)\}. \tag{9}$$

The order-two case is central to our analysis because $\mathbb{E}_Q[W_\theta^2]$ enters directly into the second-moment error of importance-weighted estimators. Its reciprocal defines the population normalized effective sample size [5, 6, 12, 17]:

$$\rho_\theta := \frac{1}{d_2(P_\theta \| Q)} = \frac{1}{\mathbb{E}_Q[W_\theta^2]} = \exp\{-D_2(P_\theta \| Q)\}. \tag{10}$$

We denote its sample analogue by $\widehat{\rho}_\theta$.

When $\mathbb{E}_Q[W_\theta^2] < \infty$, Jensen's inequality together with $\mathbb{E}_Q[W_\theta] = 1$ gives $\rho_\theta \in (0, 1]$, and $N\rho_\theta$ is the corresponding population effective sample count for a nominal sample size N. Moreover,

$$\mathbb{E}_Q[(W_\theta - 1)^2] = \rho_\theta^{-1} - 1. \tag{11}$$

Thus, normalized ESS is directly determined by the second-moment dispersion of the complete-trajectory likelihood ratio. The next section shows how this same second moment enters gradient-estimation error and, through that error, policy improvement.

4 Theoretical Analysis

The estimator formulation above leaves two questions. The first is how finite-sample gradient error affects policy improvement and how ESS enters this error for importance weighting. The second is when modifying the estimator through clipping produces a stronger update despite removing gradient contributions. We address these questions in turn.

4.1 Gradient-estimator reliability and policy improvement

We first establish the optimization consequence of gradient-estimation error. The following result applies to an arbitrary gradient estimator.

Lemma 1 (A reliable gradient estimate yields policy improvement)**.** *Let $\widehat{g}$ be an estimate of $g = \nabla J(\theta)$ satisfying $\|\widehat{g} - g\|_2 \le \varepsilon$. Suppose J is L-smooth along the update segment for $L > 0$, and define $\gamma = L^{-1}(1 - \varepsilon/\|\widehat{g}\|_2)_+$, with $\gamma = 0$ when $\widehat{g} = 0$. Then*

$$J(\theta + \gamma\widehat{g}) - J(\theta) \ge \frac{1}{2L}\left(\|\widehat{g}\|_2 - \varepsilon\right)_+^2. \tag{12}$$

Lemma 1 uses the same smoothness mechanism as existing policy-gradient analyses [8, 10]. It makes gradient-estimator reliability directly relevant to optimization: whenever the error radius is smaller than $\|\widehat{g}\|_2$, the resulting direction admits a positive one-step improvement certificate.

ESS and the IS estimator. For IS, the estimation error can be expressed explicitly in terms of the same weight second moment that defines ESS. At update k, condition on the preceding updates and write $W_k := W_{\theta_k}$, $f_k := f_{\theta_k}$, $g_k := g(\theta_k)$, $\widehat{g}_k := \widehat{g}_N(\theta_k)$, and $\rho_k := \rho_{\theta_k}$. Assume that the N rollout samples are independent draws from Q, that $d_2(P_{\theta_k}\|Q) < \infty$, and that $\mathbb{E}_Q[W_k^2\|f_k\|_2^2] < \infty$. Define the importance-weighted contribution scale

$$M_{2,k} := \frac{\mathbb{E}_Q[W_k^2\|f_k\|_2^2]}{\mathbb{E}_Q[W_k^2]} = \rho_k\mathbb{E}_Q[W_k^2\|f_k\|_2^2]. \tag{13}$$

This factorizes the weighted second moment as $M_{2,k}/\rho_k$: ρ_k^{-1} captures the concentration of the importance weights, while $M_{2,k}$ captures the magnitude of the associated score–advantage contributions under the squared-weight tilt.

Because the IS estimator is unbiased, its exact mean-squared error is

$$\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 = \frac{1}{N}\left\{\frac{M_{2,k}}{\rho_k} - \|g_k\|_2^2\right\}. \tag{14}$$

Consequently,

$$\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 \le \frac{M_{2,k}}{N\rho_k}. \tag{15}$$

Equation (14) separates the contribution of importance-weight concentration from the magnitude of the gradient contributions attached to those weights. At fixed $M_{2,k}$, decreasing ESS increases the finite-sample error through the factor ρ_k^{-1}. If f_k is bounded, then $M_{2,k} \le \|f_k\|_\infty^2$, recovering the familiar bounded-integrand form.

The MSE bound yields a high-probability error radius for the realized gradient.

Lemma 2 (Normalized ESS controls the gradient-error radius)**.** *Under the preceding setup, for any* $0 < \delta \le 1$*, with probability at least* $1 - \delta$*,*

$$\|\widehat{g}_k - g_k\|_2 \le \sqrt{\frac{M_{2,k}}{\delta N\rho_k}}. \tag{16}$$

Holding the contribution scale fixed, the reliability radius therefore grows as $(N\rho_k)^{-1/2}$. Combining this result with Lemma 1 yields the corresponding ESS-dependent policy-improvement guarantee.

Theorem 3 (ESS-dependent improvement for the IS estimator)**.** *Under the conditions of Lemma 2, suppose that* J *is* L_k*-smooth along the update segment for* $L_k > 0$*. For any* $0 < \delta \le 1$*, define*

$$\varepsilon_k(\delta) := \sqrt{\frac{M_{2,k}}{\delta N\rho_k}}, \qquad \gamma_k := \frac{1}{L_k}\left(1 - \frac{\varepsilon_k(\delta)}{\|\widehat{g}_k\|_2}\right)_+, \tag{17}$$

with $\gamma_k = 0$ *when* $\widehat{g}_k = 0$*. Then, with probability at least* $1 - \delta$*,*

$$J(\theta_k + \gamma_k\widehat{g}_k) - J(\theta_k) \ge \frac{1}{2L_k}\left(\|\widehat{g}_k\|_2 - \varepsilon_k(\delta)\right)_+^2. \tag{18}$$

Theorem 3 makes the optimization consequence of the finite-sample IS error explicit. The certificate is positive whenever

$$\|\widehat{g}_k\|_2 > \sqrt{\frac{M_{2,k}}{\delta N\rho_k}}. \tag{19}$$

Hence the usefulness of the importance-weighted update depends on the estimated gradient signal relative to its finite-sample error. Holding $M_{2,k}$ and the observed gradient magnitude fixed, decreasing ESS enlarges the error radius, contracts the admissible step, and weakens the certified improvement. This provides the finite-sample link between importance-weight concentration, gradient reliability, and policy improvement.

4.2 When does clipping improve the update?

The preceding result characterizes the reliability of importance-weighted gradient estimation. Determining whether clipping is beneficial requires a separate comparison because clipping changes the gradient estimator itself. The statistical comparison is between the sampling variation removed by the modification and the bias introduced by changing or masking gradient contributions.

Write $\widehat{g}_k(u) := \widehat{g}_N(u;\theta_k)$. For an arbitrary coefficient rule u, adding and subtracting the estimator mean gives

$$\begin{aligned} m_k(u) &:= \mathbb{E}\|\widehat{g}_k(u) - g_k\|_2^2 \\ &= \|\mathbb{E}[\widehat{g}_k(u)] - g_k\|_2^2 + \mathbb{E}\|\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]\|_2^2. \end{aligned} \tag{20}$$

The two terms are squared bias and sampling variation. Since the IS estimator is unbiased, an alternative estimator has lower MSE exactly when its reduction in sampling variation exceeds the squared bias introduced by modifying the coefficients. This gives the estimator-level bias–variance crossover.

Lower MSE alone, however, does not determine the ordering of policy improvement: modifying the estimator can also remove components aligned with the population gradient. To compare estimators rather than step-size rules, we therefore hold the optimization step fixed. The following theorem gives the corresponding expected-improvement certificate.

Theorem 4 (Improvement certificate)**.** *Suppose that J is L_k-smooth on every update segment generated by a coefficient rule u almost surely, for a constant $L_k > 0$, and that $\mathbb{E}\|\widehat{g}_k(u)\|_2^2 < \infty$. For any step size $\eta_k > 0$,*

$$\mathbb{E}\left[J(\theta_k + \eta_k \widehat{g}_k(u)) - J(\theta_k)\right] \geq \eta_k g_k^\top \mathbb{E}[\widehat{g}_k(u)] - \frac{L_k \eta_k^2}{2} \mathbb{E}\|\widehat{g}_k(u)\|_2^2. \tag{21}$$

Denote the right-hand side by

$$B_k(u; \eta_k) := \eta_k g_k^\top \mathbb{E}[\widehat{g}_k(u)] - \frac{L_k \eta_k^2}{2} \mathbb{E}\|\widehat{g}_k(u)\|_2^2. \tag{22}$$

The first term measures alignment with the population gradient, while the second is the quadratic penalty induced by smoothness. Modifying the estimator therefore improves the certificate only when the reduction in the second-moment penalty compensates for the useful gradient alignment removed by the modification.

The resulting crossover can be stated exactly.

Corollary 5 (Estimator crossover relative to IS)**.** *Let u_{IS} denote the IS coefficient rule and let u be any alternative rule. Suppose the same deterministic L_k is valid almost surely for both families of update segments. With a common step size $0 < \eta_k \leq 1/L_k$, the alternative rule has the stronger certificate if and only if*

$$B_k(u; \eta_k) > B_k(u_{\mathrm{IS}}; \eta_k) \quad \Longleftrightarrow \quad m_k(u_{\mathrm{IS}}) - m_k(u) > (1 - L_k \eta_k) \left\{\mathbb{E}\|\widehat{g}_k(u_{\mathrm{IS}})\|_2^2 - \mathbb{E}\|\widehat{g}_k(u)\|_2^2\right\}. \tag{23}$$

Since IS is unbiased, the same crossover has the equivalent alignment form

$$\|g_k\|_2^2 - g_k^\top \mathbb{E}[\widehat{g}_k(u)] < \frac{L_k \eta_k}{2} \left\{\mathbb{E}\|\widehat{g}_k(u_{\mathrm{IS}})\|_2^2 - \mathbb{E}\|\widehat{g}_k(u)\|_2^2\right\}. \tag{24}$$

The left-hand side measures the population-gradient alignment lost by modifying the estimator, while the right-hand side measures the reduction in the quadratic second-moment penalty. This is the optimization-level crossover: clipping is beneficial when the reduction in update variation is large enough to compensate for the gradient signal removed by masking.

Truncation and masking. TIS and PPO-style clipping instantiate different modifications within this comparison. TIS changes the magnitude of an importance coefficient while retaining the corresponding score contribution. PPO-style clipping instead sets selected score contributions to zero according to the advantage-dependent clipping condition. Both alter the bias and second moment of the resulting gradient estimator, and Corollary 5 compares their optimization consequences through the same alignment–second-moment criterion.

Taken together, the analysis establishes the two links needed for the remainder of the paper. For importance-weighted estimation, ESS enters the finite-sample gradient error through the second moment of the likelihood weights, and gradient reliability in turn determines the policy-improvement certificate. For clipping, the relevant comparison is whether the resulting reduction in estimator variation compensates for the population-gradient alignment removed by masking. The next section examines these predictions in controlled and large-model policy-optimization settings.

5 Policy optimization and ESS

This section makes the theoretical distinction observable in two settings. An analytically tractable Gaussian bandit first instantiates IS reliability, estimator crossover, and one-step policy improvement. Large-model trajectories then show how observed effective support evolves around delayed optimization failure.

5.1 Policy optimization in a Gaussian bandit

The Gaussian bandit makes our two-stage argument fully observable. We hold the current policy fixed and move the rollout policy farther away, which creates a monotone path from high to low normalized ESS. A quadratic reward then makes the population gradient, estimator moments, and expected one-step improvement available in closed form. We can therefore test the complete chain: ESS first predicts when IS loses finite-sample reliability, and the gain comparison then selects the estimator that makes the best use of the remaining support. Appendix A.1 gives the calculations and protocol.

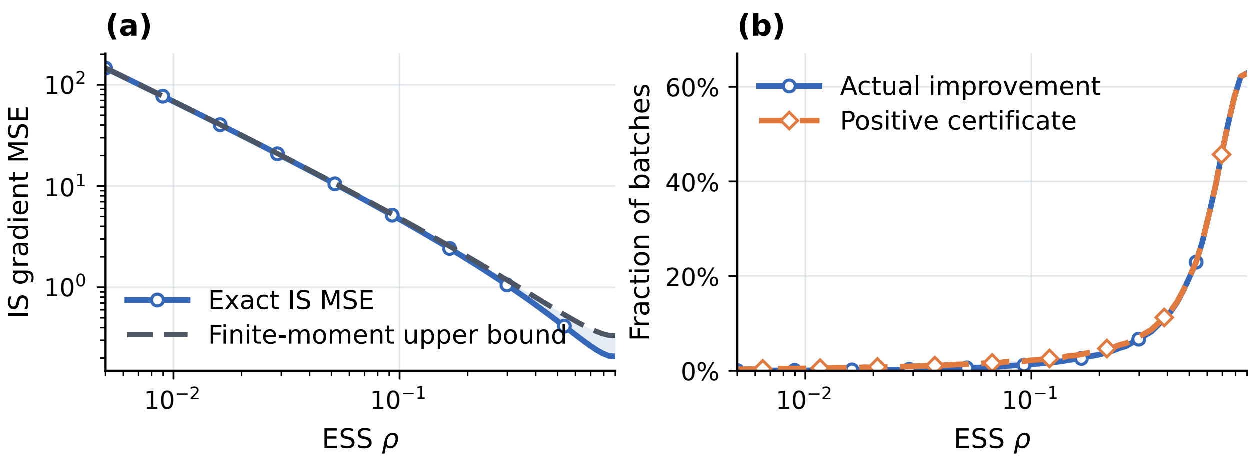

Figure 2: **I: ESS predicts IS reliability and certified improvement.** (a) Exact IS gradient MSE and the finite-moment upper bound in Equation (15). (b) Positive Theorem 3 certificates and actual positive improvement under the corresponding adaptive step.

We first ask when an IS update remains reliable. Figure 2(a) answers this question at the estimator level: the exact IS MSE stays below the bound from Lemma 2, and both rise sharply as ESS falls. Theorem 3 uses this same error radius to build an adaptive step, so estimator reliability should translate directly into policy improvement. Panel (b) confirms that translation. Across the full ESS path, the probability of actual improvement tracks the probability of a positive certificate with a maximum gap of only 0.00891. As support contracts, both probabilities fall together. ESS therefore identifies when a reliable IS update is available, but this first stage leaves one question open: which controlled estimator should be used after IS loses its advantage?

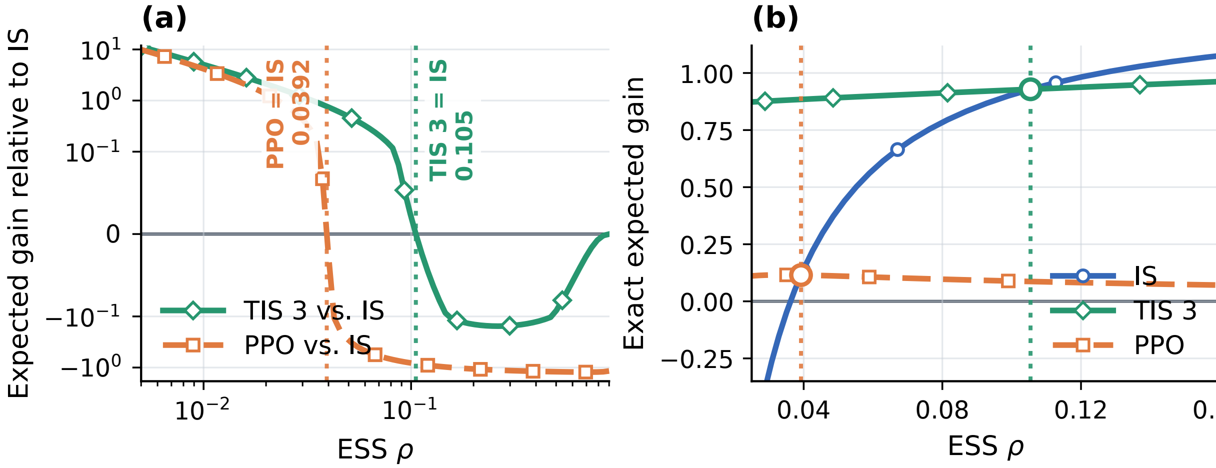

Figure 3: **II: the gain comparison selects the estimator.** We compare IS, TIS with $c = 3$, and PPO masking with $\epsilon = 0.2$ at $\eta = 0.4$. (a) Exact expected gain relative to IS; positive values favor the controlled estimator. (b) Exact expected gain around the two pairwise crossovers. Vertical lines and open markers identify the intersections.

The second Figure 3 resolves that choice through the estimator-selection criterion in Corollary 5. Figure 3(a) plots the resulting expected-gain difference relative to IS. TIS-3 has the lowest MSE throughout the path, yet IS retains more useful gradient alignment and therefore has the larger gain at high ESS. As ESS falls, the variance reduction from truncation becomes decisive, and TIS-3 overtakes IS at $\rho = 0.105$. PPO removes still more alignment, so it crosses IS only at $\rho = 0.0392$. Panel (b) makes both transitions visible as intersections of the exact policy-improvement curves. The resulting upper envelope selects IS in the high-support regime and TIS-3 in the low-support regime, exactly as predicted by the gain criterion.

Together, Figures 2 and 3 turn our theory into an operational principle. ESS determines when coefficient control becomes valuable, while the gain criterion determines which control preserves the strongest update. The remaining question is whether ESS provides this timing signal during large-model training, where the policy pair evolves after every update.

5.2 Policy optimization in large language models

The Gaussian experiment isolates the relation between ESS, gradient-estimator reliability, and policy improvement in a setting where each quantity can be evaluated directly. We now examine whether the same finite-sample mechanism is visible during large-model optimization. We consider three Qwen3-30B-A3B trajectories, consisting of an unclipped update and TIS with caps 3 and 5, and evaluate all trajectories on AIME-2024 using mean@16.

The left panel of Figure 4 places ESS within the training dynamics. All three importance-weighted updates improve substantially during the early stage of training, but their performance eventually begins to degrade. At the onset of the sustained decline, TIS-3 has ESS 0.122, whereas TIS-5 has already fallen to ESS 0.00374. The unclipped trajectory degrades later, with a ESS of 0.037 observed in training region. Thus, the delayed failure observed in the introduction occurs after a substantial deterioration in the importance-weight distribution, even though the precise ESS value at which degradation begins differs across trajectories.

The right panel tests the theoretical prediction more directly. At each evaluation checkpoint,

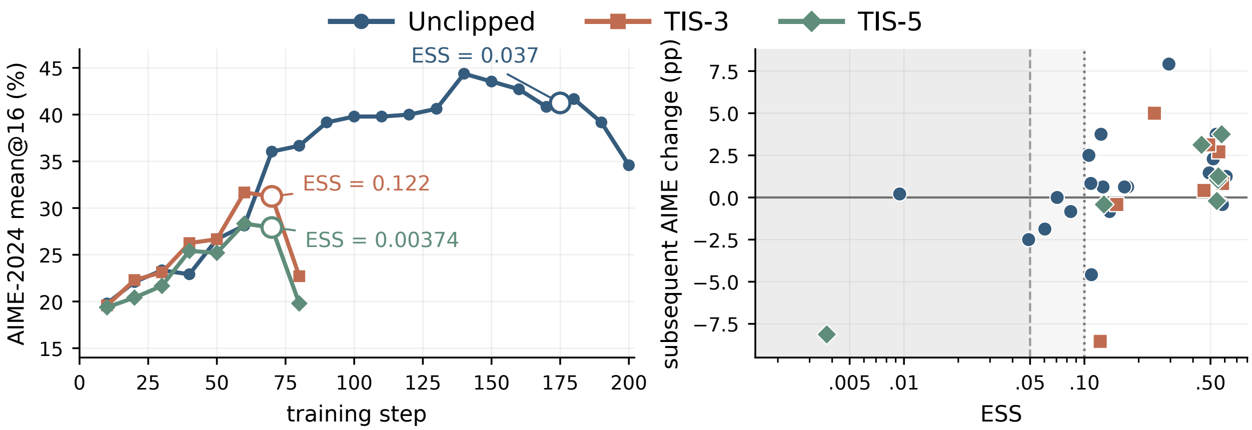

Figure 4: Left: AIME-2024 mean@16 along an unclipped trajectory and TIS with caps 3 and 5. Annotations report ESS at the onset of the sustained performance decline. Right: each point pairs ESS at an evaluation checkpoint with the change in AIME-2024 mean@16 over the following ten updates. The shaded regions indicate ESS below 0.1 and 0.05. Lower ESS is associated with substantially weaker and less reliable subsequent improvement.

we pair the current ESS with the change in AIME-2024 mean@16 over the following ten updates. The resulting pattern shows a clear change in update reliability as ESS decreases. Among intervals with ESS below 0.1, only 1 of 6 is followed by an improvement. In comparison, 20 of 27 intervals with the same ESS criterion improve. Low ESS therefore identifies a regime in which continued importance-weighted updates are substantially less likely to produce reliable improvement.

This behavior is the effect characterized in Section 4. Decreasing ESS enlarges the gradient-error radius through the factor $\rho_k^{-1/2}$ and consequently weakens the corresponding policy-improvement certificate. The large-model trajectories exhibit the same optimization consequence: as ESS deteriorates, subsequent improvement becomes less reliable. This observation motivates the optimization rule studied next: retain the importance-weighted update while its reliability remains high, and introduce clipping when ESS indicates that additional estimator control has become valuable.

6 ESS-conditioned policy optimization

The preceding experiments show that decreasing sequence ESS accompanies a loss of reliable subsequent improvement under importance-weighted updates. We now ask whether this reliability signal can be used to time clipping. Rather than applying clipping throughout training, we retain the importance-weighted update while ESS remains above a fixed threshold and activate clipping once ESS falls below it.

6.1 Update rule and protocol

Let $\widehat{g}_{\mathrm{IW}}$ denote the importance-weighted update and $\widehat{g}_{\mathrm{clip}}$ a prespecified clipped update. Given an ESS threshold τ, we use

$$\widehat{g}_{\mathrm{ESS}} = \begin{cases} \widehat{g}_{\mathrm{IW}}, & \widehat{\rho} \geq \tau, \\ \widehat{g}_{\mathrm{clip}}, & \widehat{\rho} < \tau. \end{cases} \tag{25}$$

ESS therefore determines when clipping is introduced, while the clipping rule determines how the update is modified thereafter. We use $\tau = 0.1$ in the primary experiments.

Our main experiments train Qwen3-30B-A3B for up to 200 policy updates on the boxed mathematical-reasoning data of Yu et al. [18] and evaluate AIME-2024 using mean@16. The primary importance-weighted update is TIS with ratio cap 3. At low ESS, we consider conventional GRPO clipping [15], Clip-Higher [18], and DPPO [11]. We additionally replace TIS-3 with an unclipped update to test whether the timing principle depends on truncation.

The static 30B baselines use their native token-mean aggregation, whereas the TIS-3 conditional recipes use sum normalization. We therefore interpret these comparisons as complete training recipes. The subsequent unclipped experiment keeps the importance-weighted branch fixed and isolates the effect of introducing clipping according to ESS.

6.2 Main results

Figure 5 compares ESS-conditioned optimization with static clipping across three commonly used update rules. In all three cases, delaying clipping improves both the peak and final AIME performance. Relative to the corresponding static recipes, the final-score improvements range from 6.88 to 12.3 percentage points. The common behavior is that TIS-3 preserves rapid learning early in training, while the clipping rule becomes active only after ESS enters the low-reliability regime identified in Section 5.

The collapse comparison is separated in Figure 12. Ungated TIS-3 initially improves but eventually reaches a final AIME score of 0.417%. The ESS-conditioned trajectory from Figure 5(b) instead reaches 47.3% and retains 44.8% at the end of training. Conditioning therefore retains the early learning of TIS while avoiding the late failure produced by continuing the same importance-weighted update.

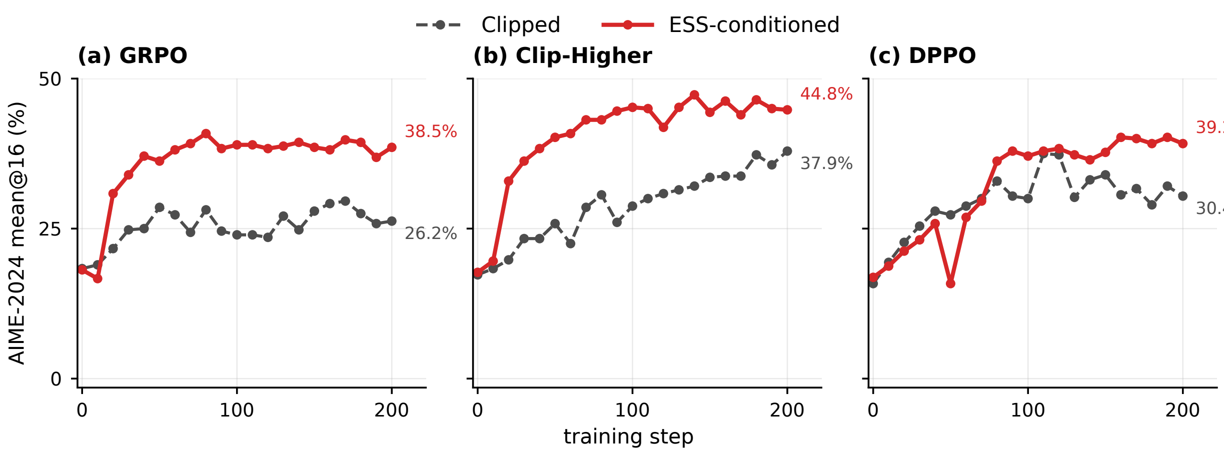

Figure 5: Qwen3-30B-A3B math RLVR over 200 policy updates. Dashed curves apply clipping throughout training. Solid curves begin with TIS-3 and switch to the corresponding clipped update when normalized sequence ESS falls below 0.1. (a) GRPO clipping. (b) Clip-Higher. (c) DPPO. End labels report the final observed AIME-2024 mean@16.

We next test whether conditioning depends on the TIS cap. Both runs start from the same unclipped update. Figure 6 leaves one run unchanged and introduces conventional clipping in the other when ESS falls below 0.1. The two trajectories attain similar peak performance, but the unclipped run subsequently degrades to 34.6%, whereas ESS-conditioned clipping retains 38.1% at step 200. The timing principle therefore extends beyond the TIS-3 branch used in the primary experiment.

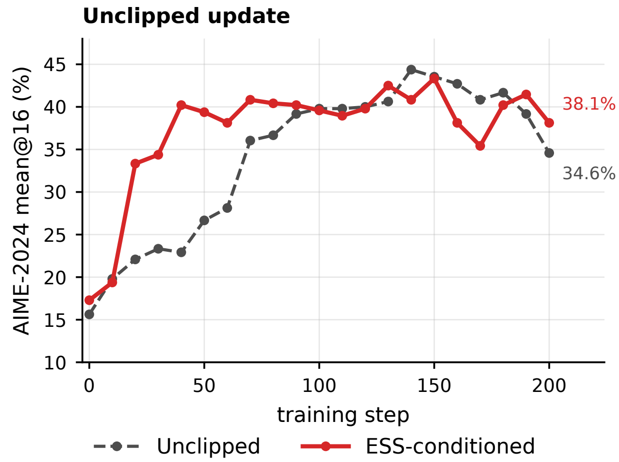

Figure 6: Qwen3-30B-A3B math RLVR with an unclipped update. Dashed: unclipped throughout. Solid: clipping below normalized sequence ESS 0.1. End labels report step-200 AIME-2024 mean@16.

6.3 Transfer across model scale and training objective

We next test whether the same construction extends beyond the Qwen3-30B-A3B setting. We first change model scale while retaining the mathematical-reasoning objective.

Figure 7 reports aggregation-matched comparisons on dense Qwen3-8B. ESS-conditioned optimization finishes above its static counterpart under both Clip-Higher and DPPO. Although the improvement is smaller than the late-stage rescue observed on Qwen3-30B-A3B, the same ordering persists across both update rules.

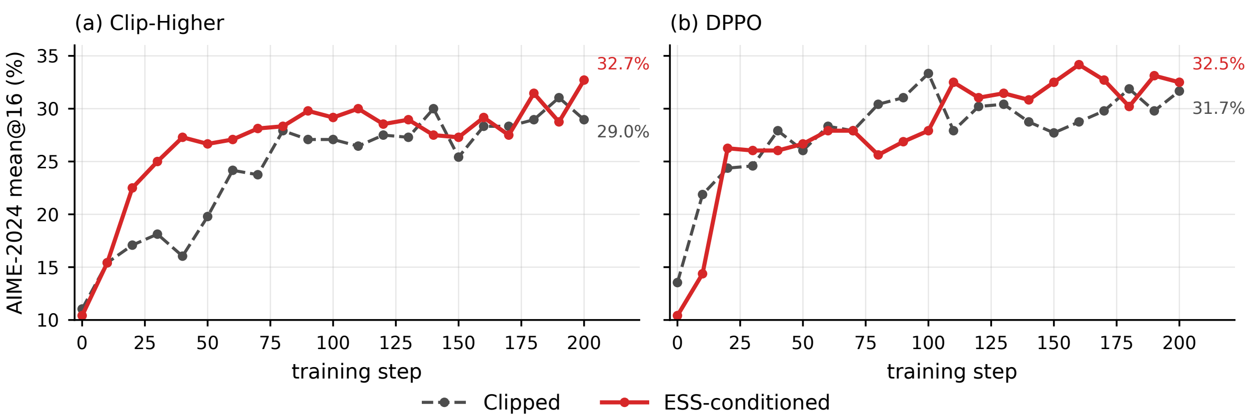

Figure 7: Qwen3-8B math RLVR over 200 policy updates. Dashed curves apply clipping throughout training, and solid curves use the ESS-conditioned counterpart. (a) Clip-Higher. (b) DPPO. End labels report the step-200 AIME-2024 mean@16.

We then reduce the model to Qwen3-1.7B while retaining the mathematical- reasoning objective. Figure 8 overlays evaluation performance and finite-rollout support for the two matched learning-rate recipes. TIS-3 + Clip reaches a final AIME score of 12.1%, compared with 10.2% for GRPO. Placing sequence ESS on the same training-step axis makes the support available to each update visible alongside its learning curve.

We further change both the objective and the data by training Qwen3-4B-Instruct-2507 on Anthropic HH-RLHF [1], using Skywork-Reward-Llama-3.1-8B [4] as the reward model. Figure 9 shows the same advantage under learned-reward optimization. The TIS-3-plus-ESS recipe reaches a final reward-model score of 60.7, compared with 43.0 for GRPO. This experiment evaluates optimization of the specified reward model rather than human preference directly.

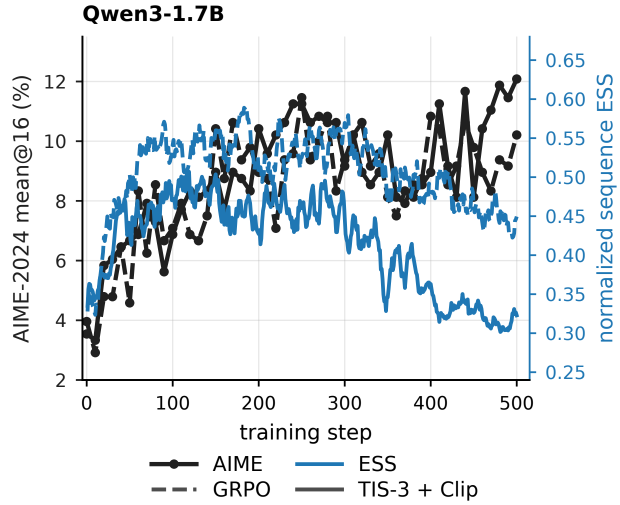

Figure 8: Qwen3-1.7B math RLVR at learning rate 10^{-6}. Black (left): AIME-2024 mean@16. Blue (right): seven-step mean normalized sequence ESS. Dashed: GRPO. Solid: TIS-3 + Clip.

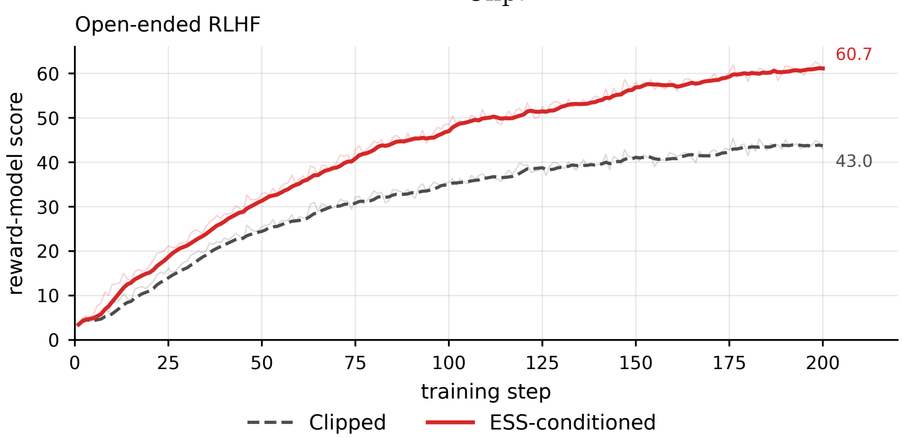

Figure 9: RLHF on Qwen3-4B-Instruct-2507 using Skywork-Reward-Llama-3.1-8B as the reward model. Thin curves show per-step reward-model scores and thick curves show their trailing seven-step means. The dashed curve is the clipped baseline and the solid curve is the ESS-conditioned update. End labels report the step-200 observations.

6.4 Ablations

The ESS-conditioned construction contains two distinct choices: the threshold determines when the update changes, while the low-ESS rule determines how it changes. We study these components separately.

ESS threshold. Figure 10 varies only τ in the TIS-3/GRPO construction. Thresholds 0.05, 0.1, and 0.2 all avoid the collapse of ungated TIS-3, with last-observed AIME scores of 43.1%, 38.5%, and 34.8%, respectively. The threshold therefore controls the timing of clipping, and the resulting performance varies with how long TIS-3 remains active. The collapse-prevention behavior persists across the tested range.

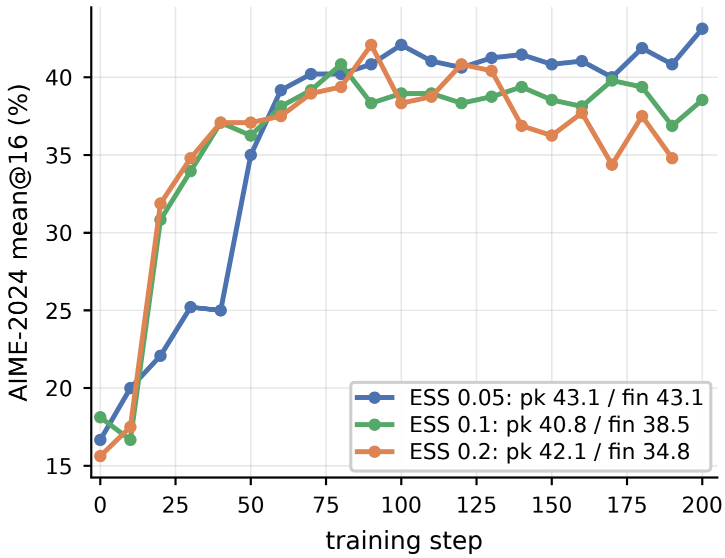

Figure 10: Sensitivity to the ESS threshold for TIS-3 with GRPO clipping on Qwen3-30B-A3B. Curves use $\tau \in \{0.05, 0.1, 0.2\}$; clipping is activated when normalized sequence ESS falls below the corresponding threshold. Legend entries report peak and final AIME-2024 mean@16.

Clipping versus skipping. ESS determines when to intervene but does not prescribe the intervention itself. We therefore replace clipping with a complete update veto while keeping the same ESS threshold.

Figure 11 isolates a complete update veto on the unclipped branch. Skipping raises the final score only slightly, from 34.6% to 35.6%, while reducing the peak from 44.4% to 35.6%. By contrast, ESS-conditioned clipping in Figure 6 reaches a 43.3% peak and retains 38.1%. The intervention must therefore preserve useful gradient signal after ESS identifies an unreliable minibatch.

The separate TIS-5 comparison in Figure 13 shows the complementary regime. Here skipping prevents complete collapse and raises the final AIME score from 0.00% to 39.4%. Together, the two comparisons separate the timing decision supplied by ESS from the estimator choice made after that transition.

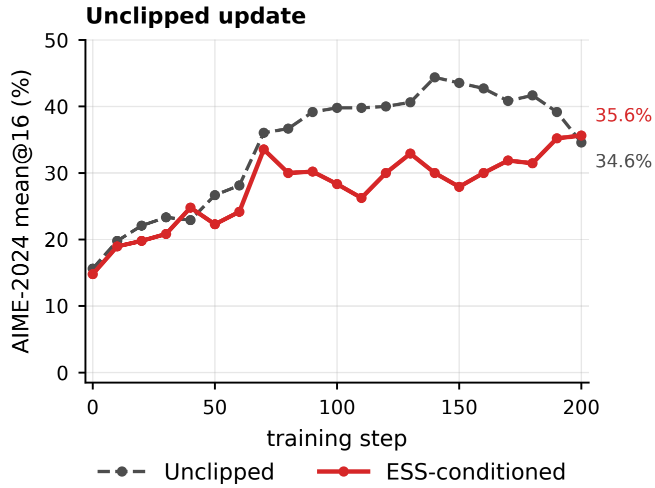

Figure 11: ESS-conditioned update skipping on the unclipped Qwen3-30B-A3B trajectory. The dashed curve applies every update, and the solid curve skips the update when normalized sequence ESS falls below 0.1. End labels report the final observed AIME-2024 mean@16.

Together, the ablations separate the two roles in ESS-conditioned optimization. The ESS threshold determines the intervention time; the post-threshold estimator determines the resulting bias–variation tradeoff. This distinction mirrors the theoretical analysis: identifying deterioration in importance-weighted reliability does not by itself determine how the gradient estimator should be modified.

7 Conclusion

We studied clipping in policy optimization through the finite-sample reliability of the gradient estimator. Off-policy policy gradients admit an importance-weighted mean-estimation formulation, under which ESS enters directly through the second moment of the importance weights. Our theory connects this finite-sample estimation error to one-step policy improvement and separately characterizes when clipping compensates for the gradient alignment it removes.

The Gaussian experiment makes these relationships explicit. As ESS decreases, the importance-weighted gradient becomes less reliable and positive policy improvement becomes less frequent. The estimator comparison further shows that lower estimation error alone is insufficient to determine the preferred update: clipping or truncation becomes advantageous only when its reduction in update variation compensates for the useful gradient signal it removes.

The language-model experiments exhibit the corresponding training behavior. Under both TIS and unclipped updates, decreasing ESS precedes a loss of reliable subsequent improvement. Using this signal to time clipping retains the rapid early learning of importance weighting while reducing its late-stage degradation. The same construction improves performance across clipping rules, model scales, and both RLVR and learned-reward optimization.

ESS therefore serves a specific role in policy optimization: it characterizes the finite-sample reliability of importance-weighted estimation and provides a principled signal for deciding when additional estimator control becomes valuable. Clipping remains an estimator choice; ESS determines when that choice matters.

A Numerical Experiments

A.1 Gaussian bandit: exact calculations

This appendix gives the calculations underlying the Gaussian experiment in Section 5.1. All estimator moments and expected one-step gains are available from truncated Gaussian moments. Monte Carlo simulation is used only to evaluate the batch-level probabilities in Figure 2(b).

Gaussian policy pair and ESS path. Fix the current policy

$$P_\theta = \mathcal{N}(\theta, 1) \tag{26}$$

and consider rollout policies

$$Q_\Delta = \mathcal{N}(\theta - \Delta, 1), \qquad \Delta \geq 0. \tag{27}$$

Thus, increasing Δ changes only the distance between the rollout and current policies. Let

$$z := a - \theta.$$

Then

$$z \sim \mathcal{N}(0, 1) \quad \text{under } P_\theta, \qquad z \sim \mathcal{N}(-\Delta, 1) \quad \text{under } Q_\Delta.$$

The likelihood ratio is

$$W_\Delta(z) = \frac{dP_\theta}{dQ_\Delta}(z) = \exp\left\{\Delta z + \frac{\Delta^2}{2}\right\}. \tag{28}$$

A direct calculation gives

$$\mathbb{E}_{Q_\Delta}[W_\Delta^2] = e^{\Delta^2}, \qquad \rho(\Delta) = e^{-\Delta^2}. \tag{29}$$

Hence Δ and normalized ESS are in one-to-one correspondence:

$$\Delta = \sqrt{-\log \rho}. \tag{30}$$

All figures are therefore parameterized by ρ, while the calculations below are carried out through the corresponding Δ.

Reward, population gradient, and per-sample contribution. We use the quadratic reward

$$R(a) = -\frac{1}{2}(a - a^\star)^2 \tag{31}$$

and define

$$d := a^\star - \theta > 0. \tag{32}$$

The current-policy objective and gradient are

$$J(\theta) = -\frac{1}{2}\left\{d^2 + 1\right\}, \qquad \nabla J(\theta) = d. \tag{33}$$

We hold θ, and therefore d, fixed while varying Δ.

Using the exact current-policy value as the baseline, the advantage of a sample with centered action z is

$$A(z) = dz + \frac{1 - z^2}{2}. \tag{34}$$

Since the score of $\mathcal{N}(\theta, 1)$ is z, the corresponding policy-gradient contribution is the cubic polynomial

$$\psi(z) := zA(z) = dz^2 + \frac{z - z^3}{2}. \tag{35}$$

In particular,

$$\mathbb{E}_{P_\theta}[\psi(z)] = d,$$

as required by the policy-gradient identity.

We compare three estimators using samples $z_1, \ldots, z_N \overset{\text{iid}}{\sim} \mathcal{N}(-\Delta, 1)$. Their per-sample contributions are

$$\xi_{\text{IS}}(z) := W_\Delta(z)\psi(z), \tag{36}$$

$$\xi_{\text{T}}(z) := \min\{W_\Delta(z), c\}\psi(z), \tag{37}$$

$$\xi_{\text{P}}(z) := W_\Delta(z)\psi(z)M_{\epsilon,\Delta}(z), \tag{38}$$

where $c = 3$, $\epsilon = 0.2$, and

$$M_{\epsilon,\Delta}(z) = \mathbf{1}\{A(z) \geq 0,\ W_\Delta(z) \leq 1 + \epsilon\} + \mathbf{1}\{A(z) < 0,\ W_\Delta(z) \geq 1 - \epsilon\}. \tag{39}$$

The corresponding batch estimator is

$$\widehat{g}_u = \frac{1}{N}\sum_{i=1}^{N} \xi_u(z_i), \qquad u \in \{\text{IS}, \text{T}, \text{P}\}. \tag{40}$$

This representation reduces every quantity used in the figures to the first two moments of ξ_u. Define

$$\mu_u(\Delta) := \mathbb{E}_{Q_\Delta}[\xi_u(z)], \qquad \nu_u(\Delta) := \mathbb{E}_{Q_\Delta}[\xi_u(z)^2]. \tag{41}$$

Then

$$\mathbb{E}[\widehat{g}_u] = \mu_u, \tag{42}$$

$$\mathbb{E}[\widehat{g}_u^2] = \mu_u^2 + \frac{\nu_u - \mu_u^2}{N}, \tag{43}$$

$$\mathbb{E}[(\widehat{g}_u - d)^2] = (\mu_u - d)^2 + \frac{\nu_u - \mu_u^2}{N}. \tag{44}$$

Thus, once (μ_u, ν_u) are known, both the estimator MSE and the second-moment term in the policy-improvement certificate are known exactly.

Exact Gaussian moment calculation. The required moments can be evaluated by change of measure. For IS,

$$\mu_{\text{IS}} = d, \qquad \nu_{\text{IS}} = e^{\Delta^2}\mathbb{E}_{V\sim\mathcal{N}(\Delta,1)}[\psi(V)^2]. \tag{45}$$

For PPO masking,

$$\mu_{\text{P}} = \mathbb{E}_{U\sim\mathcal{N}(0,1)}[\psi(U)M_{\epsilon,\Delta}(U)], \tag{46}$$

$$\nu_{\text{P}} = e^{\Delta^2}\mathbb{E}_{V\sim\mathcal{N}(\Delta,1)}[\psi(V)^2 M_{\epsilon,\Delta}(V)]. \tag{47}$$

For TIS, the truncation boundary is obtained from $W_\Delta(z) = c$:

$$b_c(\Delta) := \frac{\log c - \Delta^2/2}{\Delta}. \tag{48}$$

Splitting the integral at this boundary gives

$$\mu_{\mathrm{T}} = \mathbb{E}_{U\sim\mathcal{N}(0,1)}[\psi(U)\mathbf{1}\{U \le b_c\}] + c\,\mathbb{E}_{U\sim\mathcal{N}(-\Delta,1)}[\psi(U)\mathbf{1}\{U > b_c\}], \tag{49}$$

$$\nu_{\mathrm{T}} = e^{\Delta^2}\mathbb{E}_{V\sim\mathcal{N}(\Delta,1)}[\psi(V)^2\mathbf{1}\{V \le b_c\}] + c^2\,\mathbb{E}_{V\sim\mathcal{N}(-\Delta,1)}[\psi(V)^2\mathbf{1}\{V > b_c\}]. \tag{50}$$

These expectations have closed forms because ψ is a cubic polynomial and all estimator boundaries are linear thresholds in z. In particular, the advantage changes sign at

$$z = d \pm \sqrt{d^2+1}, \tag{51}$$

while the PPO ratio boundaries are

$$b_{\pm}(\Delta) = \frac{\log(1\pm\epsilon) - \Delta^2/2}{\Delta}. \tag{52}$$

The real line can therefore be partitioned at the two advantage roots and the two ratio boundaries, on each of which the PPO mask is constant.

For completeness, let

$$I_j(a,b) := \int_a^b z^j \phi(z)\,dz. \tag{53}$$

The truncated standard-normal moments satisfy

$$I_j(a,b) = a^{j-1}\phi(a) - b^{j-1}\phi(b) + (j-1)I_{j-2}(a,b), \qquad j \ge 2, \tag{54}$$

with

$$I_0(a,b) = \Phi(b) - \Phi(a), \qquad I_1(a,b) = \phi(a) - \phi(b). \tag{55}$$

After shifting the integration variable for nonzero Gaussian means, Equations (45)–(50) reduce to finite sums of these moments. All curves described below, except the batch probabilities in Figure 2(b), are evaluated from these exact moments.

Figure 2: ESS and IS reliability. We first reproduce the quantities used to validate the ESS-dependent reliability result. For IS, define

$$M_2(\Delta) := \rho(\Delta)\nu_{\mathrm{IS}}(\Delta) = \mathbb{E}_{V\sim\mathcal{N}(\Delta,1)}[\psi(V)^2]. \tag{56}$$

For the setting used in the figure, $d = 2$, this becomes

$$M_2(\Delta) = \frac{\Delta^6}{4} - 2\Delta^5 + \frac{29\Delta^4}{4} - 18\Delta^3 + \frac{65\Delta^2}{2} - 24\Delta + \frac{29}{2}. \tag{57}$$

Because IS is unbiased, Equation (44) gives

$$\mathrm{MSE}_{\mathrm{IS}}(\Delta) = \frac{1}{N}\left\{\frac{M_2(\Delta)}{\rho(\Delta)} - d^2\right\}. \tag{58}$$

The finite-moment bound from Equation (15) is

$$\mathrm{MSE}_{\mathrm{IS}}(\Delta) \le \frac{M_2(\Delta)}{N\rho(\Delta)}. \tag{59}$$

Figure 2(a) plots these two quantities against $\rho = e^{-\Delta^2}$. Thus, every point in the left plot is obtained directly from Equations (58) and (59); no Monte Carlo approximation is used.

Figure 2(b) evaluates the corresponding batch-level policy-improvement statement. We use

$$N = 32, \qquad d = 2, \qquad \delta = 0.1,$$

and evaluate 81 ESS values logarithmically spaced between 0.005 and 0.9. For each ρ, we set

$$\Delta = \sqrt{-\log \rho}$$

and compute the theoretical error radius

$$\varepsilon(\rho) = \sqrt{\frac{M_2(\Delta)}{\delta N \rho}}. \tag{60}$$

At each ESS value, we independently generate 10^5 batches of size N from Q_Δ. For every batch, we compute

$$\widehat{g}_{\mathrm{IS}} = \frac{1}{N}\sum_{i=1}^{N} W_\Delta(z_i)\psi(z_i) \tag{61}$$

and the adaptive step from Theorem 3,

$$\gamma = \left(1 - \frac{\varepsilon(\rho)}{|\widehat{g}_{\mathrm{IS}}|}\right)_+, \tag{62}$$

where $L = 1$. The theorem gives a positive certificate exactly when

$$|\widehat{g}_{\mathrm{IS}}| > \varepsilon(\rho). \tag{63}$$

The exact reward change produced by the same batch is

$$\Delta J = d\gamma\widehat{g}_{\mathrm{IS}} - \frac{\gamma^2}{2}\widehat{g}_{\mathrm{IS}}^2. \tag{64}$$

Figure 2(b) reports, at each ESS value, the fraction of batches satisfying Equation (63) and the fraction satisfying $\Delta J > 0$. These are respectively the "positive certificate" and "actual improvement" curves in the figure.

As an additional check of the high-probability result, we also record the event

$$|\widehat{g}_{\mathrm{IS}} - d| \le \varepsilon(\rho). \tag{65}$$

Across the displayed ESS grid, its minimum empirical coverage is 0.98277, which exceeds the required $1 - \delta = 0.9$.

Figure 3: estimator crossover. We next compare IS, TIS-3, and PPO at a common fixed step $\eta = 0.4$. For the quadratic objective, the smoothness inequality is an equality. Indeed, for any realized scalar update h,

$$J(\theta + \eta h) - J(\theta) = \eta d h - \frac{\eta^2}{2}h^2. \tag{66}$$

Taking expectation with $h = \widehat{g}_u$ gives the exact expected gain

$$\mathcal{G}_u(\eta) := \mathbb{E}[J(\theta + \eta\widehat{g}_u) - J(\theta)] = \eta d\mu_u - \frac{\eta^2}{2}\left(\mu_u^2 + \frac{\nu_u - \mu_u^2}{N}\right). \tag{67}$$

Thus, the fixed-step certificate in Theorem 4 is exact in this experiment.

Figure 3(a) plots

$$\mathcal{G}_{\mathrm{T}}(\eta) - \mathcal{G}_{\mathrm{IS}}(\eta) \qquad \text{and} \qquad \mathcal{G}_{\mathrm{P}}(\eta) - \mathcal{G}_{\mathrm{IS}}(\eta) \tag{68}$$

against ESS. Positive values therefore indicate that the corresponding modified estimator has larger expected one-step improvement than IS.

These differences are exactly equivalent to the crossover criterion in Corollary 5. Let

$$S_u := \mathbb{E}[\widehat{g}_u^2], \qquad m_u := \mathbb{E}[(\widehat{g}_u - d)^2]. \tag{69}$$

Since $L = 1$,

$$\{m_{\mathrm{IS}} - m_u\} - (1-\eta)\{S_{\mathrm{IS}} - S_u\} = \frac{2}{\eta}\left\{\mathcal{G}_u(\eta) - \mathcal{G}_{\mathrm{IS}}(\eta)\right\}. \tag{70}$$

Consequently, the zero crossings in Figure 3(a) are precisely the estimator crossovers predicted by the corollary.

Using the exact moments above and solving these one-dimensional equalities gives

$$\rho_{\mathrm{IS,TIS3}} = 0.105, \qquad \rho_{\mathrm{IS,PPO}} = 0.0392. \tag{71}$$

Figure 3(b) plots the three exact gains $\mathcal{G}_{\mathrm{IS}}$, $\mathcal{G}_{\mathrm{T}}$, and $\mathcal{G}_{\mathrm{P}}$ in the neighborhood of these two crossover points. The vertical lines and open markers indicate the same roots reported in Equation (71).

TIS-3 has the smallest gradient MSE throughout the displayed ESS range, but this does not make it the preferred update everywhere. At high ESS, IS retains the larger expected gain. TIS-3 overtakes IS only after ρ falls below 0.105. PPO crosses IS later, at $\rho = 0.0392$, and remains below TIS-3 throughout the displayed range. The experiment therefore illustrates the distinction established by the theory: estimator MSE and policy improvement need not induce the same ordering.

B Additional coefficient-rule details

The main text treats coefficient control generically because different rules alter different parts of the gradient. A factorwise detached upper truncation uses $u_{\mathrm{T},t}(\mathbf{r}, A) := \min\{r_t, c\}$. Its contribution obeys

$$\|\psi_k(u_{\mathrm{T}}; Z)\|_2 \le c|A(Z)| \sum_{t=1}^{T} \|\ell_{\theta_k,t}(Z)\|_2, \tag{72}$$

so it has a direct second-moment envelope whenever the right-hand side has a finite second moment.

PPO masking is different. For $A \ge 0$ and $c = 1 + \epsilon$,

$$0 \le r - \min\{r, c\} = (r-c)_+ \le r\mathbf{1}\{r > c\} = r - rM_\epsilon(r, A), \tag{73}$$

where $M_\epsilon(r, A) = \mathbf{1}\{A \ge 0,\ r \le 1+\epsilon\} + \mathbf{1}\{A < 0,\ r \ge 1-\epsilon\}$. Truncation retains a bounded positive coefficient above the cap, whereas PPO removes that contribution. On the negative-advantage lower branch, PPO masks small ratios whereas upper truncation does not. A complete-sample mask is another distinct rule: it sets $u_t(\mathbf{r}, A) = W_k M_\epsilon(W_k, A)$ for every factor. We state which rule is used in each experiment rather than treating these implementations as interchangeable.

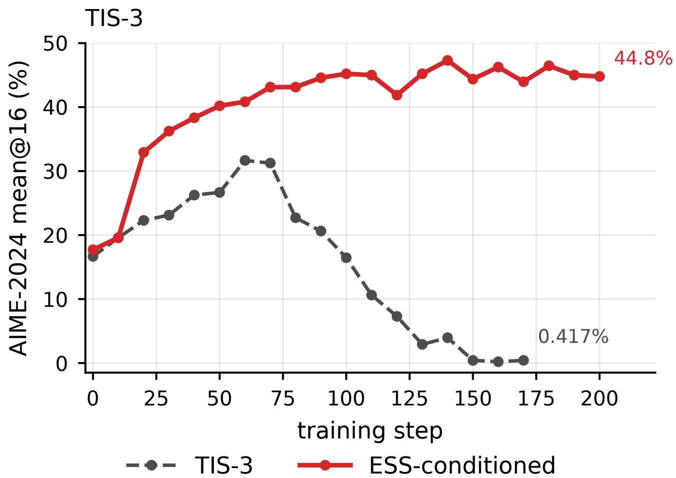

Figure 12: Separated TIS-3 collapse comparison from the main 30B experiment. Ungated TIS-3 is shown against the ESS-conditioned Clip-Higher trajectory used in Figure 5(b).

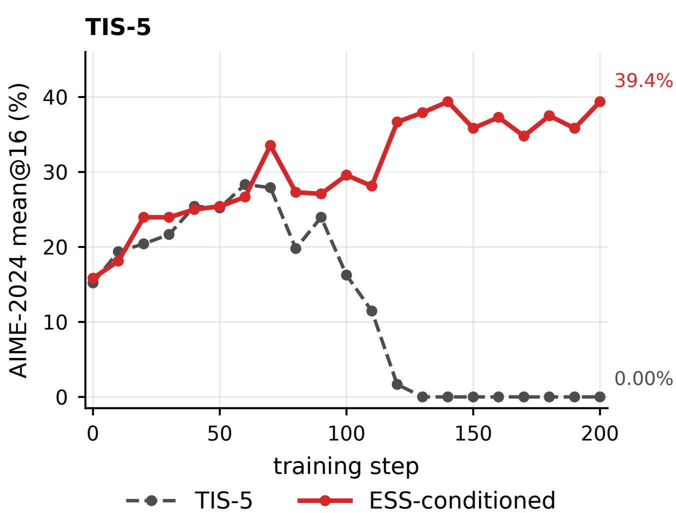

Figure 13: Separated TIS-5 skipping comparison on Qwen3-30B-A3B. The dashed curve continues TIS-5 throughout training; the solid curve skips low-ESS updates.

C Additional RLVR diagnostics

C.1 Intervention and training dynamics

The main text reports evaluation performance. We record here how often the conditional rule intervenes and the accompanying training dynamics.

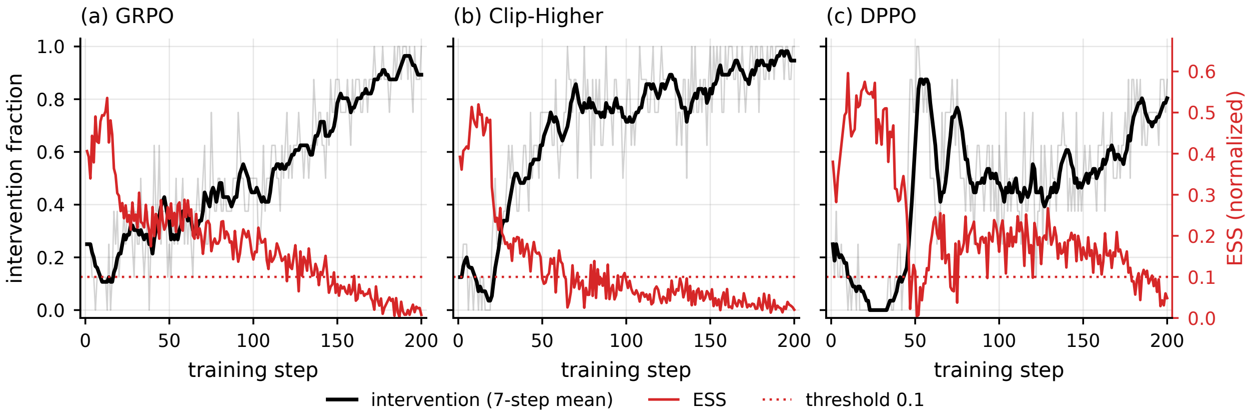

Figure 14: Realized ESS-conditioned intervention schedules. Each panel pairs the per-step intervention fraction (thin black) and its trailing seven-step mean (thick black) with normalized ESS (red). GRPO and DPPO use raw ESS; Clip-Higher uses shaped ESS after the coefficient cap. The dotted line marks the threshold 0.1.

Figure 14 shows the intended temporal ordering: intervention is already possible early, but its frequency increases as ESS contracts. The remaining figures record response length, entropy, and reward diagnostics.

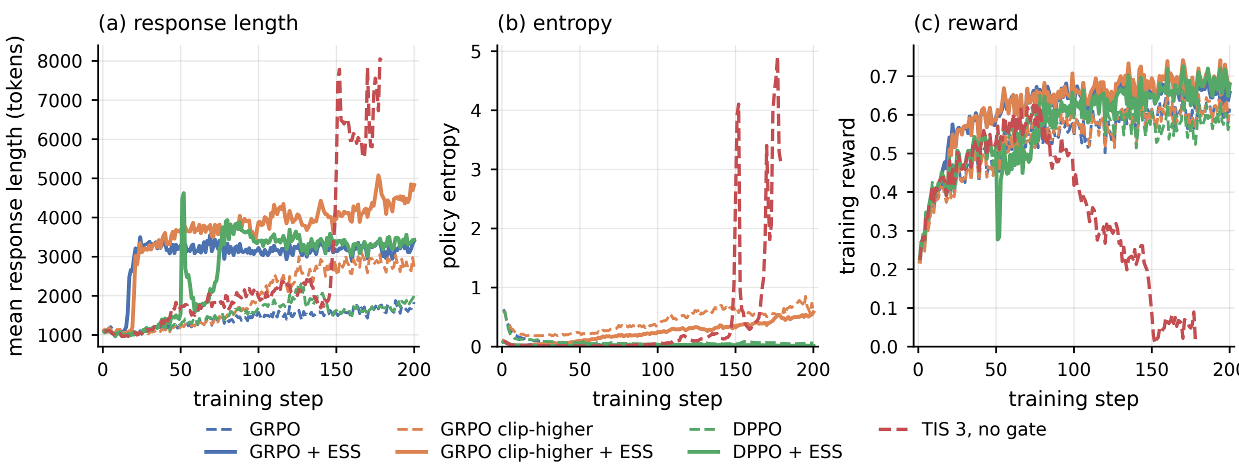

Figure 15: Training dynamics for the main comparisons. The static TIS 3 run develops extreme response length and entropy together with a collapse in training reward. The safeguards and their ESS-conditional variants avoid this failure over the reported 200 steps.

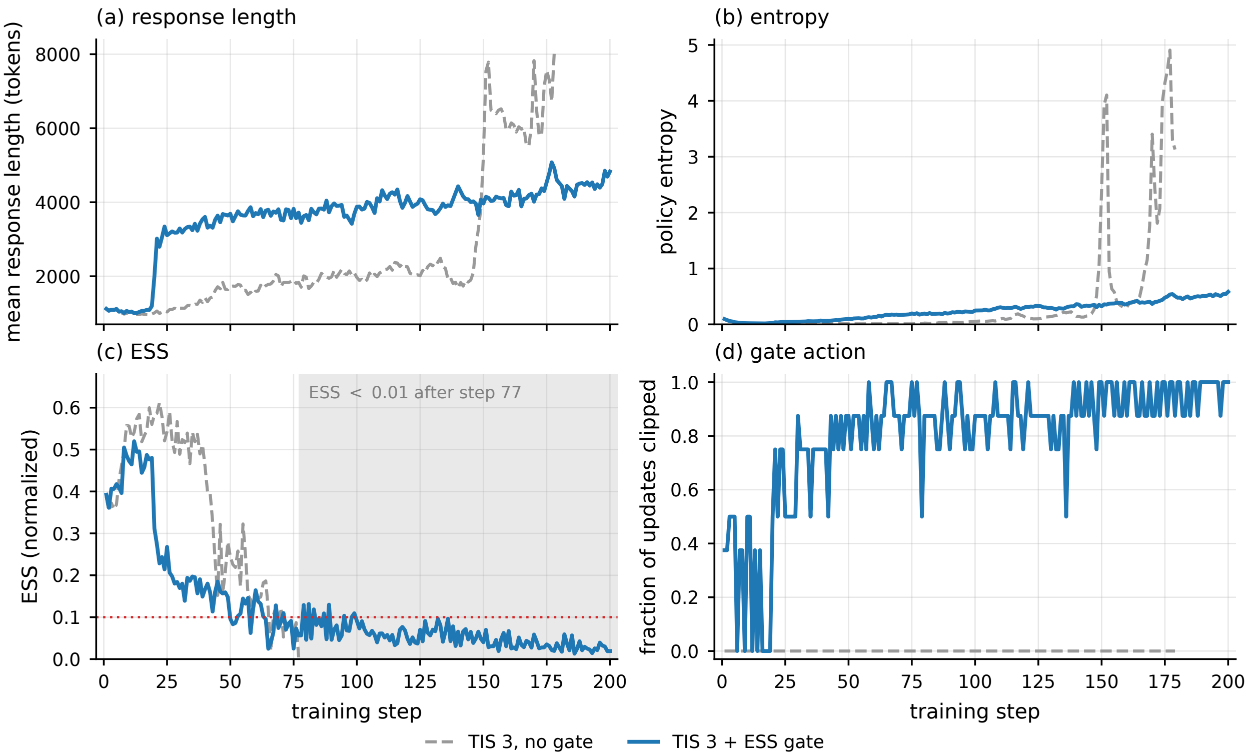

Figure 16: Detailed TIS 3 comparison. Under static TIS, ESS falls below 0.01 after step 77 and the response length and entropy diverge. The fixed-0.1 intervention increases the clipped fraction as ESS declines and keeps both diagnostics controlled.

D Proofs for the theoretical results

At update k, all expectations in this appendix are taken after conditioning on the preceding updates. The current policy is therefore fixed and the unused sampling units are independent draws from Q. We suppress this conditioning throughout.

D.1 Policy-gradient identity and mean-square bound

By definition, $f_k = (R - b(X))\nabla_\theta \log P_{\theta_k}(Y \mid X)$ and $W_k = dP_{\theta_k}/dQ$. Change of measure gives

$$\begin{aligned}
\mathbb{E}_Q[W_k f_k] &= \mathbb{E}_{P_{\theta_k}}[(R - b(X))\nabla_\theta \log P_{\theta_k}(Y \mid X)] \\
&= \mathbb{E}_{P_{\theta_k}}[R\nabla_\theta \log P_{\theta_k}(Y \mid X)] \\
&\quad - \mathbb{E}_{P_{\theta_k}}[b(X)\nabla_\theta \log P_{\theta_k}(Y \mid X)].
\end{aligned} \tag{74}$$

For the baseline term, condition further on X. Under the standard regularity condition that permits differentiation under the conditional integral,

$$\begin{aligned}
&\mathbb{E}_{P_{\theta_k}}[b(X)\nabla_\theta \log P_{\theta_k}(Y \mid X) \mid X] \\
&\quad = b(X)\int P_{\theta_k}(y \mid X)\nabla_\theta \log P_{\theta_k}(y \mid X)\, d\lambda(y) \\
&\quad = b(X)\nabla_\theta \int P_{\theta_k}(y \mid X)\, d\lambda(y) \\
&\quad = b(X)\nabla_\theta 1 = 0.
\end{aligned} \tag{75}$$

The remaining term is the score-function identity. Hence

$$\mathbb{E}_Q[W_k f_k] = \nabla J(\theta_k) = g_k. \tag{76}$$

Averaging N independent copies gives $\mathbb{E}[\widehat{g}_k] = g_k$.

For the mean-square calculation, define $\xi_{k,i} := W_{k,i} f_{k,i} - g_k$. Equation (76) implies $\mathbb{E}[\xi_{k,i}] = 0$. Since $\widehat{g}_k - g_k = N^{-1}\sum_{i=1}^N \xi_{k,i}$,

$$\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 = \frac{1}{N^2}\sum_{i=1}^N \mathbb{E}\|\xi_{k,i}\|_2^2 + \frac{1}{N^2}\sum_{i\neq j}\mathbb{E}[\xi_{k,i}^\top \xi_{k,j}]. \tag{77}$$

For $i \neq j$, independence and zero means give

$$\mathbb{E}[\xi_{k,i}^\top \xi_{k,j}] = \mathbb{E}[\xi_{k,i}]^\top \mathbb{E}[\xi_{k,j}] = 0. \tag{78}$$

The diagonal terms are identical, so

$$\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 = \frac{1}{N}\mathbb{E}_Q\|W_k f_k - g_k\|_2^2. \tag{79}$$

Expanding the remaining squared norm yields

$$\begin{aligned}
\mathbb{E}_Q\|W_k f_k - g_k\|_2^2 &= \mathbb{E}_Q[W_k^2\|f_k\|_2^2] - 2g_k^\top \mathbb{E}_Q[W_k f_k] + \|g_k\|_2^2 \\
&= \mathbb{E}_Q[W_k^2\|f_k\|_2^2] - \|g_k\|_2^2,
\end{aligned} \tag{80}$$

where the last equality uses Equation (76). Combining Equations (79) and (80) gives

$$\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 = \frac{1}{N}\left\{\mathbb{E}_Q[W_k^2\|f_k\|_2^2] - \|g_k\|_2^2\right\}. \tag{81}$$

Finally, Equation (13) implies

$$\begin{aligned}\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 &\le \frac{1}{N}\mathbb{E}_Q[W_k^2\|f_k\|_2^2] \\ &= \frac{M_{2,k}}{N\rho_k},\end{aligned} \tag{82}$$

which proves Equation (15).

D.2 Proof of Lemma 2

If $M_{2,k} = 0$, Equation (15) makes the claim immediate. Otherwise, apply Markov's inequality to the nonnegative random variable $\|\widehat{g}_k - g_k\|_2^2$:

$$\begin{aligned}&\Pr\left(\|\widehat{g}_k - g_k\|_2 > \sqrt{\frac{M_{2,k}}{\delta N\rho_k}}\right) \\ &= \Pr\left(\|\widehat{g}_k - g_k\|_2^2 > \frac{M_{2,k}}{\delta N\rho_k}\right) \\ &\le \frac{\mathbb{E}\|\widehat{g}_k - g_k\|_2^2}{M_{2,k}/(\delta N\rho_k)} \\ &\le \delta,\end{aligned} \tag{83}$$

where the last line uses Equation (15). Taking complements proves Equation (16).

D.3 Proof of Lemma 1

Let $\|\widehat{g} - g\|_2 \le \varepsilon$. Smoothness gives, for every $\gamma \ge 0$,

$$J(\theta + \gamma\widehat{g}) - J(\theta) \ge \gamma g^\top \widehat{g} - \frac{L\gamma^2}{2}\|\widehat{g}\|_2^2. \tag{84}$$

The inner product satisfies

$$\begin{aligned}g^\top\widehat{g} &= \|\widehat{g}\|_2^2 + (g - \widehat{g})^\top\widehat{g} \\ &\ge \|\widehat{g}\|_2^2 - \|g - \widehat{g}\|_2\|\widehat{g}\|_2 \\ &\ge \|\widehat{g}\|_2(\|\widehat{g}\|_2 - \varepsilon).\end{aligned} \tag{85}$$

Substituting Equation (85) into Equation (84) gives

$$J(\theta + \gamma\widehat{g}) - J(\theta) \ge \gamma\|\widehat{g}\|_2(\|\widehat{g}\|_2 - \varepsilon) - \frac{L\gamma^2}{2}\|\widehat{g}\|_2^2. \tag{86}$$

For $\widehat{g} \neq 0$, the derivative of the right-hand side is

$$\|\widehat{g}\|_2(\|\widehat{g}\|_2 - \varepsilon) - L\gamma\|\widehat{g}\|_2^2. \tag{87}$$

Its nonnegative maximizer is $\gamma = L^{-1}(1 - \varepsilon/\|\widehat{g}\|_2)_+$. When $\widehat{g} = 0$, both the chosen step and the claimed lower bound are zero. Substituting the maximizer into Equation (86) proves Equation (12).

D.4 Proof of Theorem 3

Lemma 2 shows that the event

$$\|\widehat{g}_k - g_k\|_2 \leq \sqrt{\frac{M_{2,k}}{\delta N \rho_k}} \tag{88}$$

has probability at least $1-\delta$. On this event, apply Lemma 1 with $\theta = \theta_k$, $\widehat{g} = \widehat{g}_k$, $\varepsilon = \sqrt{M_{2,k}/(\delta N \rho_k)}$, and $L = L_k$. The resulting inequality is exactly Equation (18).

D.5 Risk and certificate comparisons for coefficient rules

For a coefficient rule u, add and subtract the estimator mean:

$$\widehat{g}_k(u) - g_k = \{\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]\} + \{\mathbb{E}[\widehat{g}_k(u)] - g_k\}\,. \tag{89}$$

Squaring and taking expectation gives

$$\begin{aligned} m_k(u) &= \mathbb{E}\,\|\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]\|_2^2 + \|\mathbb{E}[\widehat{g}_k(u)] - g_k\|_2^2 \\ &\quad + 2\{\mathbb{E}[\widehat{g}_k(u)] - g_k\}^\top \mathbb{E}\left[\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]\right] \\ &= \mathbb{E}\,\|\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]\|_2^2 + \|\mathbb{E}[\widehat{g}_k(u)] - g_k\|_2^2, \end{aligned} \tag{90}$$

because the centered term has mean zero. This proves Equation (20). For an iid sample mean, the first term in Equation (90) is the per-sample variation divided by N.

It remains to connect total risk to the expected policy-improvement certificate. For any random estimate $\widehat{g}$ of g_k, smoothness gives

$$J(\theta_k + \eta_k \widehat{g}) - J(\theta_k) \geq \eta_k g_k^\top \widehat{g} - \frac{L_k \eta_k^2}{2}\|\widehat{g}\|_2^2. \tag{91}$$

Taking expectation in Equation (91), with $\mathbb{E}\|\widehat{g}\|_2^2 < \infty$, gives

$$\mathbb{E}[J(\theta_k + \eta_k \widehat{g}) - J(\theta_k)] \geq \eta_k g_k^\top \mathbb{E}[\widehat{g}] - \frac{L_k \eta_k^2}{2}\mathbb{E}\|\widehat{g}\|_2^2. \tag{92}$$

Setting $\widehat{g} = \widehat{g}_k(u)$ proves Theorem 4. The polarization identity, followed by expectation, gives

$$2 g_k^\top \mathbb{E}[\widehat{g}] = \|g_k\|_2^2 + \mathbb{E}\|\widehat{g}\|_2^2 - \mathbb{E}\|\widehat{g} - g_k\|_2^2. \tag{93}$$

Substituting Equation (93) into the certificate and subtracting the IS certificate gives

$$\begin{aligned} B_k(u;\eta_k) - B_k(u_{\mathrm{IS}};\eta_k) &= \frac{\eta_k}{2}\left[m_k(u_{\mathrm{IS}}) - m_k(u)\right] \\ &\quad - \frac{\eta_k}{2}(1 - L_k\eta_k)\left\{\mathbb{E}\|\widehat{g}_k(u_{\mathrm{IS}})\|_2^2 - \mathbb{E}\|\widehat{g}_k(u)\|_2^2\right\}. \end{aligned} \tag{94}$$

Since $\eta_k > 0$, the right-hand side is positive exactly under Equation (23), proving Corollary 5. Since the IS estimator is unbiased, subtracting the certificate definitions instead gives

$$\begin{aligned} B_k(u;\eta_k) - B_k(u_{\mathrm{IS}};\eta_k) &= -\eta_k\left\{\|g_k\|_2^2 - g_k^\top \mathbb{E}[\widehat{g}_k(u)]\right\} \\ &\quad + \frac{L_k\eta_k^2}{2}\left\{\mathbb{E}\|\widehat{g}_k(u_{\mathrm{IS}})\|_2^2 - \mathbb{E}\|\widehat{g}_k(u)\|_2^2\right\}. \end{aligned} \tag{95}$$

Its positivity is equivalent to Equation (24).

D.6 Intact prompt groups

Suppose M prompts are independent and each prompt has G conditionally independent responses. Keep each prompt group intact and define its contribution by $\Xi_{k,i} := G^{-1}\sum_{j=1}^{G} W_{k,ij} f_{k,ij}$. A leave-one-out baseline is admissible because it excludes the focal response. Conditional on the prompt and the remaining responses, the focal score still has mean zero. Hence $\mathbb{E}[\Xi_{k,i}] = g_k$.

Jensen's inequality gives

$$
\begin{aligned}
\|\Xi_{k,i}\|_2^2 &= \left\| \frac{1}{G} \sum_{j=1}^{G} W_{k,ij} f_{k,ij} \right\|_2^2 \\
&\le \frac{1}{G} \sum_{j=1}^{G} W_{k,ij}^2 \|f_{k,ij}\|_2^2. \qquad (96)
\end{aligned}
$$

Repeating the sample-mean calculation across the M independent prompt groups gives

$$
\begin{aligned}
\mathbb{E} \left\| \frac{1}{M} \sum_{i=1}^{M} \Xi_{k,i} - g_k \right\|_2^2 &\le \frac{1}{M} \mathbb{E}\|\Xi_{k,1}\|_2^2 \\
&\le \frac{\|f_k\|_\infty^2}{M\rho_k}. \qquad (97)
\end{aligned}
$$

Markov's inequality then gives the bounded-contribution specialization of Lemma 2 with the number of independent prompts M in place of the number of responses N.

D.7 Bounded-contribution specialization

If $\|f_k\|_\infty < \infty$, then $M_{2,k} \le \|f_k\|_\infty^2$. Lemma 2 therefore recovers the conventional radius $\|f_k\|_\infty / \sqrt{\delta N \rho_k}$ as a direct special case.

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